Course Introduction

CSCI-P 556

ZORAN TIGANJ

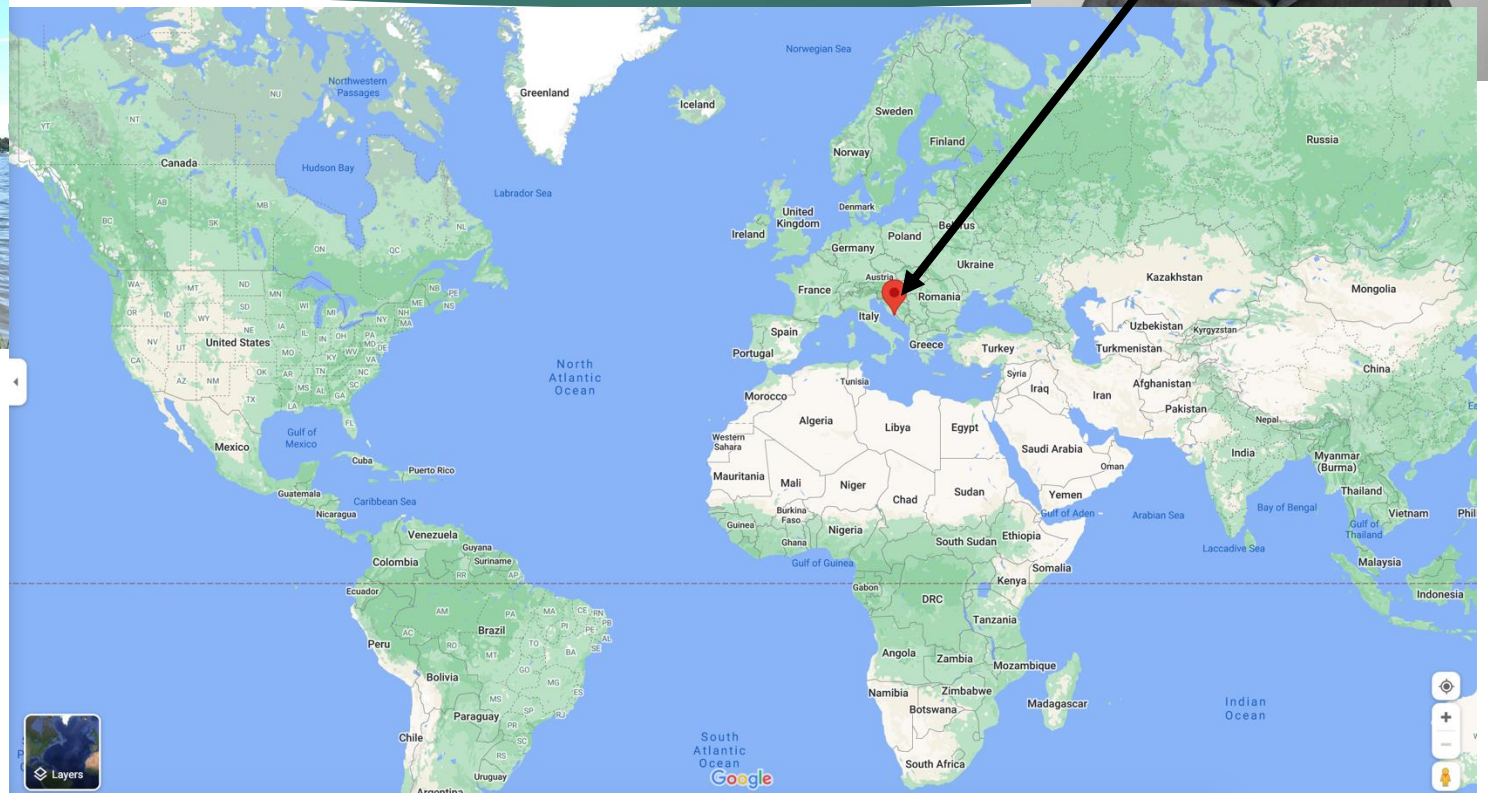
Today

- Instructor introduction
- What is machine learning?
- Course structure, objectives and requirements
- Zoran's research

Instructor

- **Zoran Tiganj**
  - Joined IU, in January of 2020
  - Assistant Professor
  - Computer Science, School of Informatics, Computing and Engineering (SICE)
    - Also affiliated with Department of Psychological and Brain Sciences
    - Program in Cognitive Science
    - Program in Neuroscience
    - Program in Data science
  - Previous:
    - Postdoc/Research Scientist at Boston University, Department of Psychological and Brain Sciences 2013-2019
    - Postdoc and University of Versailles, France
    - PhD in Computer Science from INRIA, France, 2011
    - MS in Electrical Engineering from University of Zagreb, Croatia, 2008

Instructor


Norwegian Sea
Greenland
Iceland
Northwestern Passages
Canada
Hudson Bay
Labrador Sea
Sweden
Finland
Norway
Russia
United Kingdom
Denmark
Ireland
Poland
Germany
Ukraine
France
Austria
Romania
Italy
Kazakhstan
Mongolia
Uzbekistan
Kyrgyzstan
Turkmenistan
Spain
Portugal
Greece
Turkey
Syria
Iraq
Iran
Afghanistan
Pakistan
China
Nepal
India
Myanmar (Burma)
Thailand
Vietnam
Malaysia
Indonesia
United States
North Atlantic Ocean
Tunisia
Morocco
Algeria
Libya
Egypt
Western Sahara
Mauritania
Mali
Niger
Chad
Sudan
Saudi Arabia
Oman
Yemen
Gulf of Aden
Arabian Sea
Bay of Bengal
Gulf of Thailand
Laccadive Sea
Burkina Faso
Guinea
Ghana
Nigeria
South Sudan
Ethiopia
Somalia
Kenya
Gulf of Guinea
Gabon
DRC
Tanzania
Angola
Zambia
Mozambique
Namibia
Zimbabwe
Botswana
Madagascar
South Africa
Indian Ocean
South Atlantic Ocean
Gulf of Mexico
Mexico
Cuba
Puerto Rico
Caribbean Sea
Guatemala
Nicaragua
Venezuela
Guyana
Suriname
Colombia
Ecuador
Peru
Brazil
Bolivia
Paraguay
Chile
Uruguay
Layers
Google

Instructor

- **Zoran Tiganj**
  - Homepage: https://homes.luddy.indiana.edu/ztiganj
  - Email: ztiganj@iu.edu
  - Office hours: Thursday 4pm-5pm and by appointment (follow Cavas for more info and any changes)
  - Main research interests:
    - Cognitively inspired Artificial Intelligence
    - Machine learning applied on neural (brain) recordings

Associate Instructors

- Md Rysul Kabir (mdrkabir@iu.edu)
- Sara Zomorodi (szomorod@iu.edu)
- Eugene Melnyk (ymelnyk@iu.edu)
- Billy Dickson (dicksonb@iu.edu)

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What is Machine Learning?

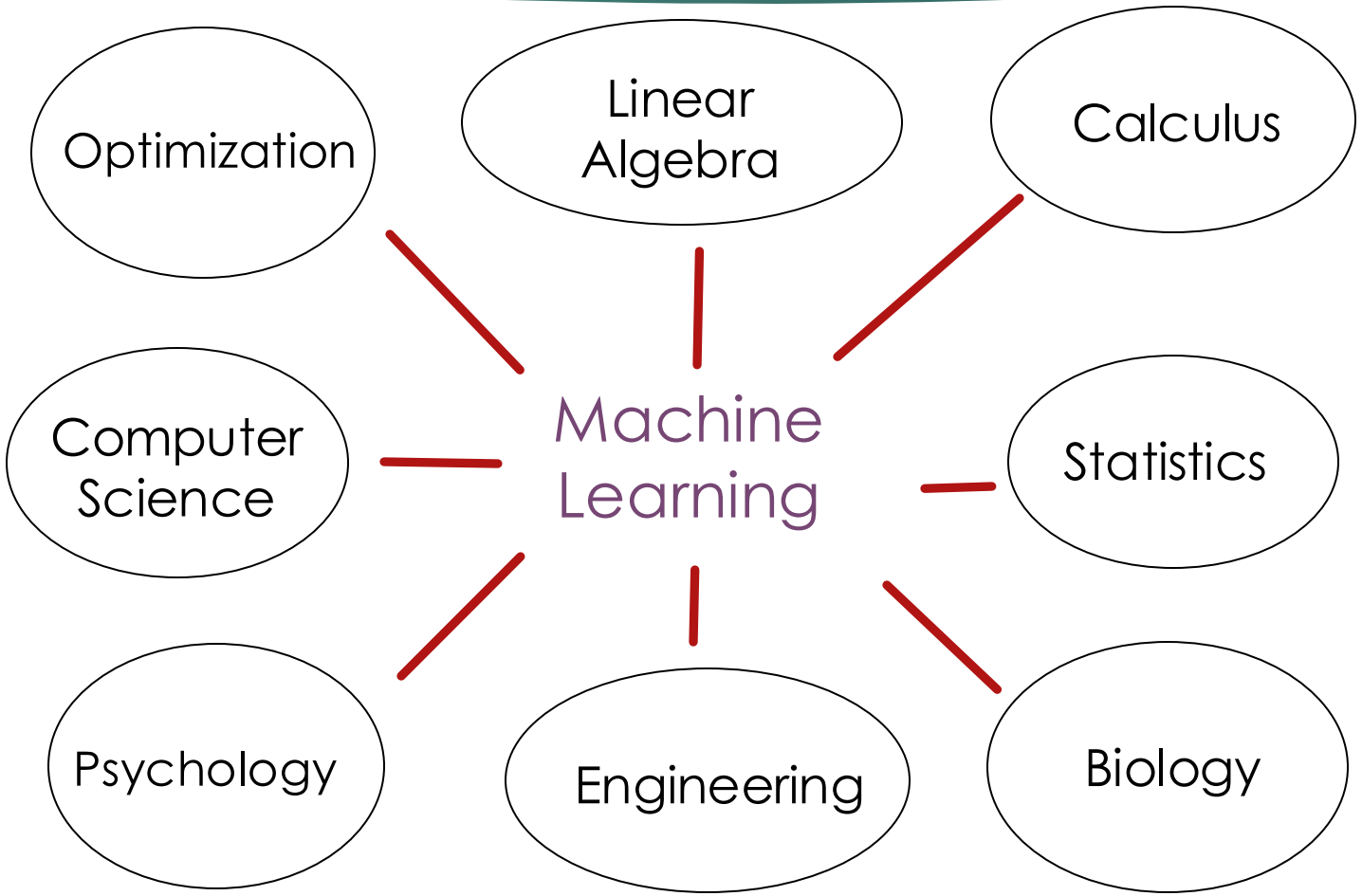
- Definition:
  - "A set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision making under uncertainty" ~ Kevin P. Murphy

What is Machine Learning?

- ICML 2024 Topics of interest include (but are not limited to):
- General Machine Learning (active learning, clustering, online learning, ranking, reinforcement learning, supervised, semi- and self-supervised learning, time series analysis, etc.)
- Deep Learning (architectures, generative models, deep reinforcement learning, etc.)
- Learning Theory (bandits, game theory, statistical learning theory, etc.)
- Optimization (convex and non-convex optimization, matrix/tensor methods, stochastic, online, non-smooth, composite, etc.)
- Probabilistic Inference (Bayesian methods, graphical models, Monte Carlo methods, etc.)
- Trustworthy Machine Learning (accountability, causality, fairness, privacy, robustness, etc.)
- Applications (computational biology, crowdsourcing, healthcare, neuroscience, social good, climate science, etc.)

https://icml.cc/Conferences/2023/CallForPapers

Related Disciplines



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List of Topics Covered

The main aim of the course is to provide skills to apply machine learning algorithms on real applications. Topics that we will cover include:

- *Introduction to Machine Learning*
- *Linear regression*
- *Logistic regression*
- *Support vector machines*
- *Decision trees*
- *Dimensionality reduction*
- *Gaussian mixtures*
- *Introduction to deep neural networks*
- *Convolutional neural networks*
- *Recurrent neural networks*
- *Autoencoders*
- Transformers
- *Generative adversarial networks*
- *Reinforcement learning*

Applications

- Applications will include:
  - time-series prediction,
  - classification,
  - data visualization,
  - computer vision,
  - natural language processing,
  - robotics.

Production-ready Python frameworks

- We will be using production-ready Python frameworks, primarily:
  - **Scikit-Learn**
    - Implements many Machine Learning algorithms efficiently
    - Very easy to use
  - **TensorFlow**
    - Useful for running very large neural networks efficiently by using GPUs (graphics processing unit)
    - TensorFlow was created at Google and it was open sourced in November 2015.
  - **Keras**
    - High-level Deep Learning API that makes it very simple to train and run neural networks.
    - TensorFlow comes with its own implementation of this API, called tf.keras.

What do I Expect and Assume?

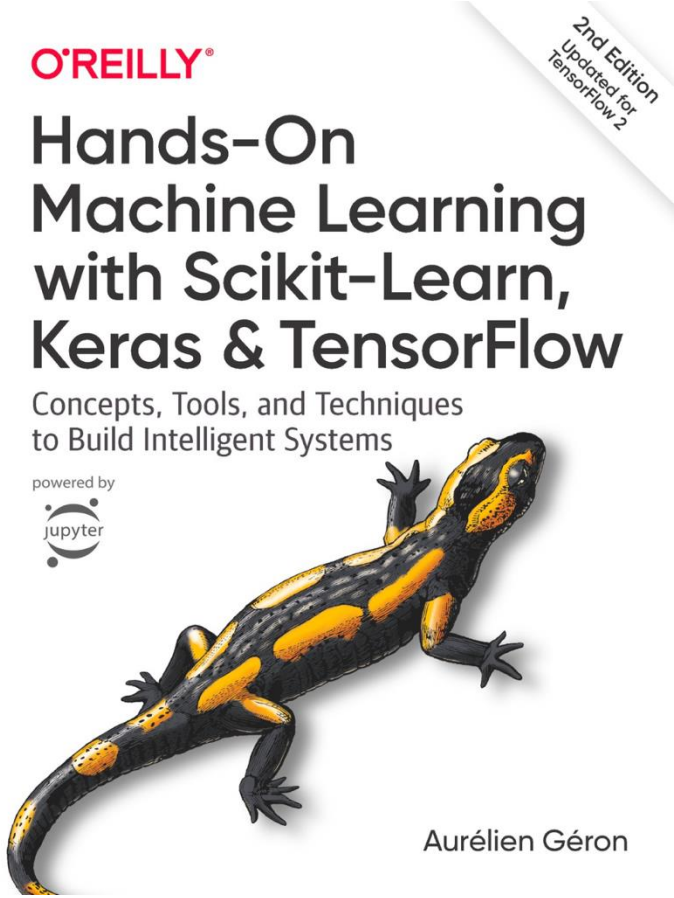
- Basic mathematical skills
  - Probability
  - Calculus (derivatives/integrals)
  - Linear algebra basics
- You are patient and hardworking
  - You will attend the lectures
  - You will review the slides
  - You will read course material
- Your integrity is impeccable
- You are motivated to learn

What Should you Expect?

- I am here to help you learn
  - I am always happy to answer your questions
- I will treat you with respect and listen thoughtfully to your questions
- Any feedback is always appreciated

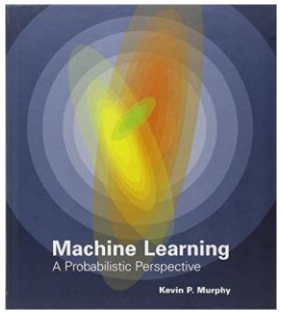
Textbook Information

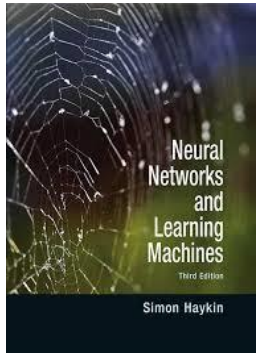
- The lectures will primarily follow **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow** book by Aurélien Géron (2nd or 3rd edition is recommended).
  - An electronic or e-book copy is fine for the purposes of the class.
  - You will not need the book for the exams, so it is practical to share a book with other students in the class.
  - Schedule, readings, and resources will be available via Canvas
  - Here you can find all of the code used in the book: https://github.com/ageron/handson-ml3



Textbook Information

- Recommended Material
  - *Pattern Recognition and Machine Learning*, by C. M. Bishop, Springer 2006 (free copy available online)
  - *Machine Learning: A Probabilistic Perspective*, by Kevin Murphy, MIT Press 2012
  - *Neural Networks and Learning Machines: Third Edition*, by Simon Haykin, Prentice Hall, 2009
  - Deep Learning by Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016





Grading Policy

- *Assignments (homework)* (30%), total of 4 assignments (two questions each), most of which will require substantial programming projects. The assignments can be done individually or in groups of two, three, four or five. Your lowest graded assignment will be dropped.
- *Quizzes, attendance and in-class activities* (20%). These are not meant to be stressful events, but ways to give you practice understanding course concepts while rewarding regular class participation and attendance. They also give instructor feedback on how well the class is progressing. Your lowest graded quiz will be dropped.
- *Midterm exam* (15%), in class. If you have an unavoidable conflict with the midterm day or time, please notify us at least one week before the midterm date.
- *Final exam* (35%), in class. If you have an unavoidable conflict with the final day or time, please notify us at least one week before the final date.

Please make any grade disputes within 2 weeks

Late submission policy

- Assignments and quizzes will be accepted after the due date, but with 5% late penalty per day of delay
  - e.g., if you are 4 days late, you will receive 20% less points than what you would receive if you submitted the assignment or quiz on time.
  - All assignments and quizzes will be submitted online through Canvas. The late penalty is applied automatically through Canvas and we will make no exceptions for late submissions (if we do, it would make grading unfair to students who submitted their assignments on time) – Canvas will count a day penalty even if you are late by less than a minute.
  - We understand that delays can sometimes happen and that is why we drop your lowest scored assignment question and quiz.

Attendance policy:

- Students are expected to attend all class sessions.
- We will use Quizzes, TopHat and sometimes signup sheets to determine class attendance.
- Students attending more than 90% of the classes will earn maximum attendance points, students attending between 80% and 90% of classes will earn 90% of attendance points etc.
- If you are sick/traveling, you can watch recordings on Canvas (no need to notify us).

Grading Scale

- **A+:** 97% - 100%
- **B+:** 87% - 89%
- **C+:** 77% - 79%
- **D+:** 67% - 69%
- **F**: 0 – 59%

A: 93%-96%

B: 83%-86%

C: 73%-76%

D: 63%-66%

A-: 90%-92%

B-: 80%-82%

C-: 70%-72%

D-: 60%-62%

We might adjust the grading scale, but only in your favor.

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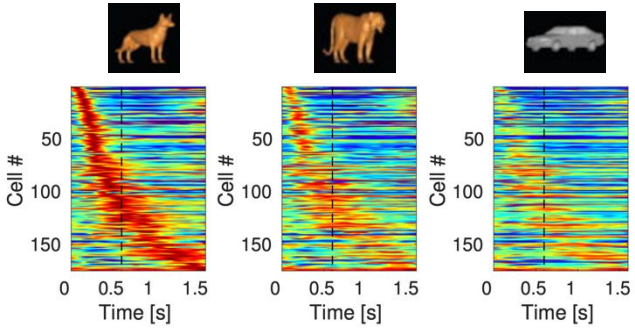
Zoran's research

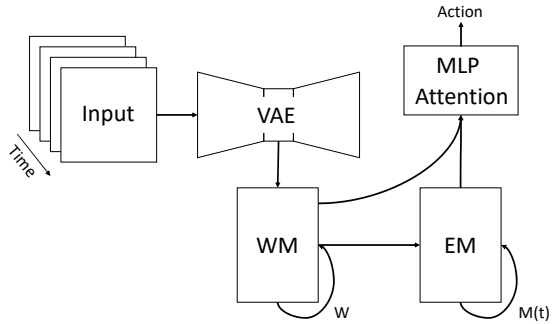
Electrophysiological recordings to test mechanistic hypotheses

Constructing AI systems capable of lifelong self-supervised learning

Neuroscience

Artificial Intelligence

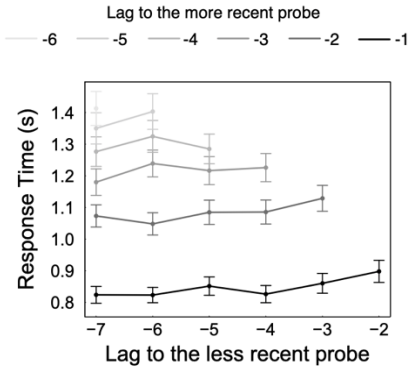


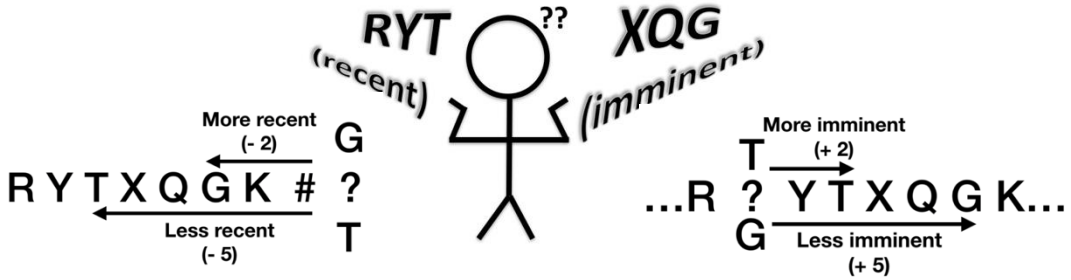


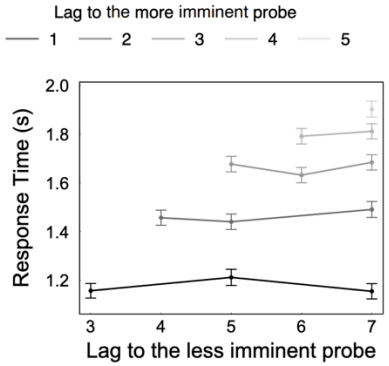
Mental maps

Cognitive Science

Behavioral data provide insights into how people store memory and how they use it to predict the future



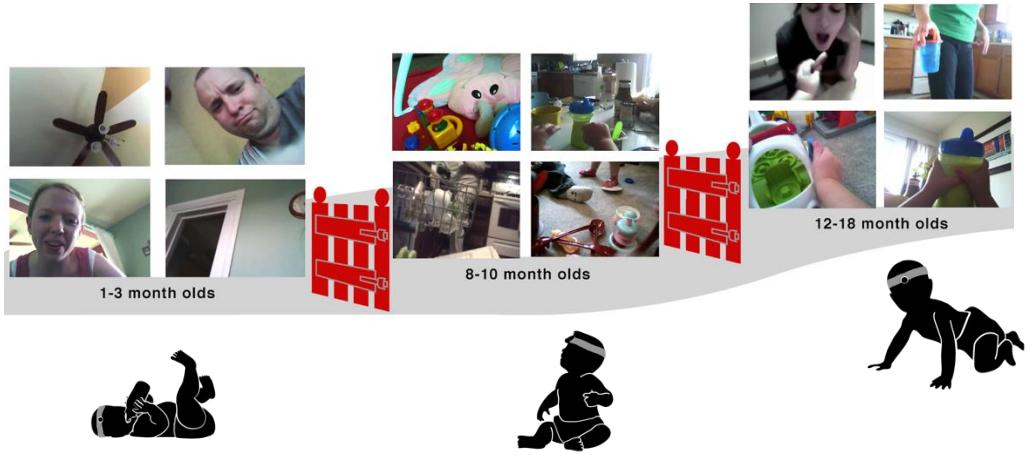




https://homes.luddy.indiana.edu/ztiganj

Curriculum learning with infant egocentric videos

- Early visual learning in humans coincides with changes in motor development that create visual experiences with fundamentally different visual properties.
- **Motivation**: Are there benefits to learning from such developmentally structured datasets?
- **What we did**: Tested computationally the outcome of learning from infant videos, with various orderings.



Saber Sheybani, Himanshu Hansaria, Justin Wood, Linda Smith, Zoran Tiganj.
Curriculum learning with infant egocentric videos. NeurIPS (Spotlight paper), 2023.

Curriculum learning with infant egocentric videos

- We computationally tested the impact of infant curriculum on learning

Homeview dataset: over 500 hours of head camera videos collected by 101 infants from 2 to 24 months of age.

See https://cogdev.sitehost.iu.edu/ for more info about the dataset.

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Curriculum learning with infant egocentric videos

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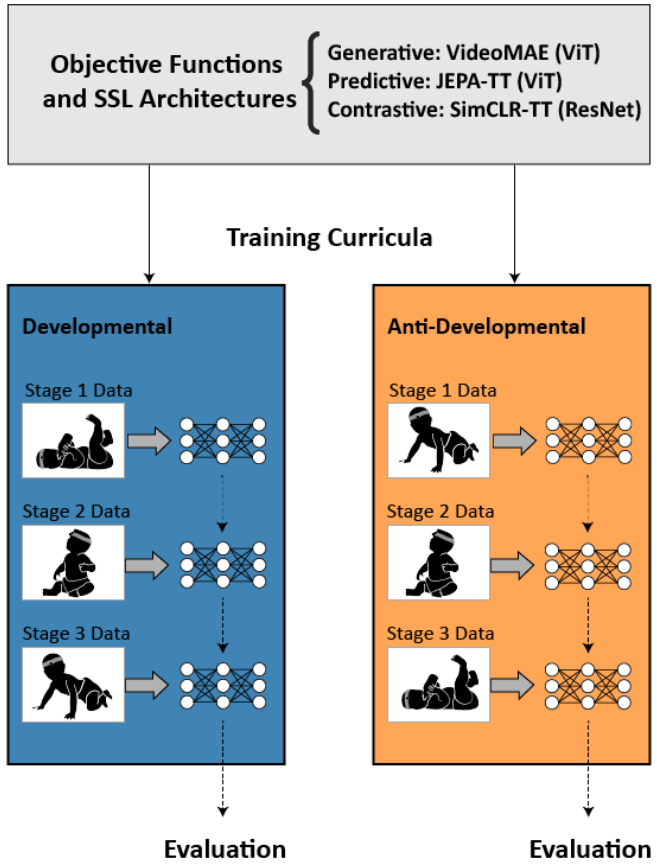

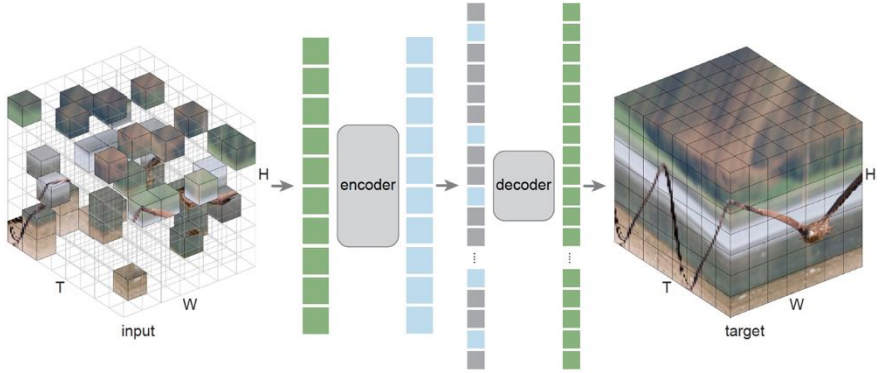

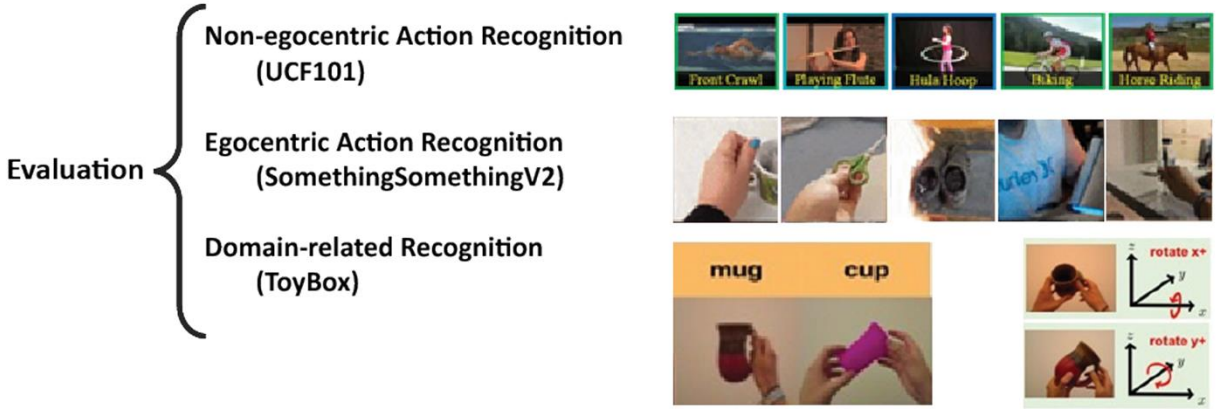

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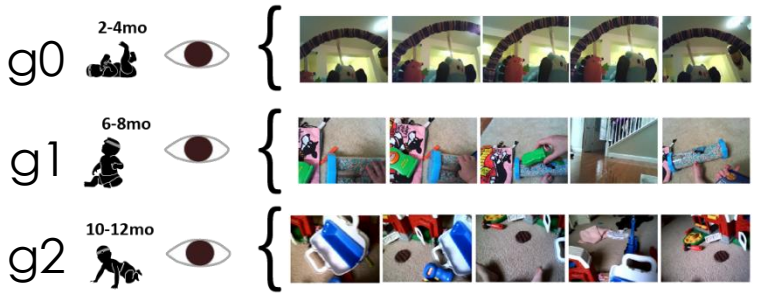
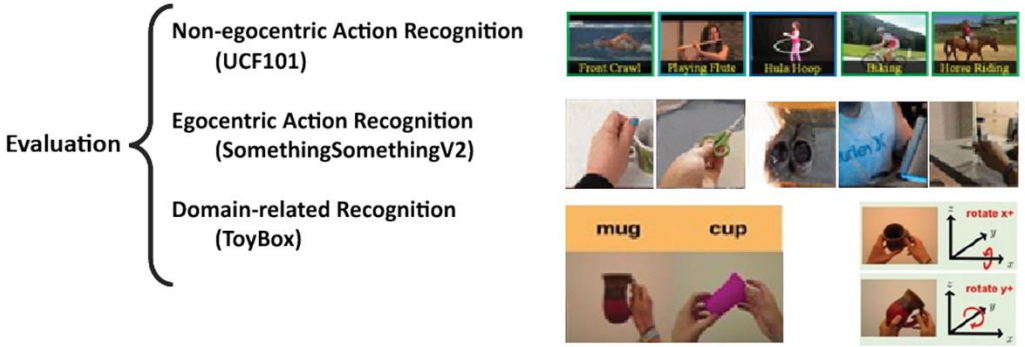
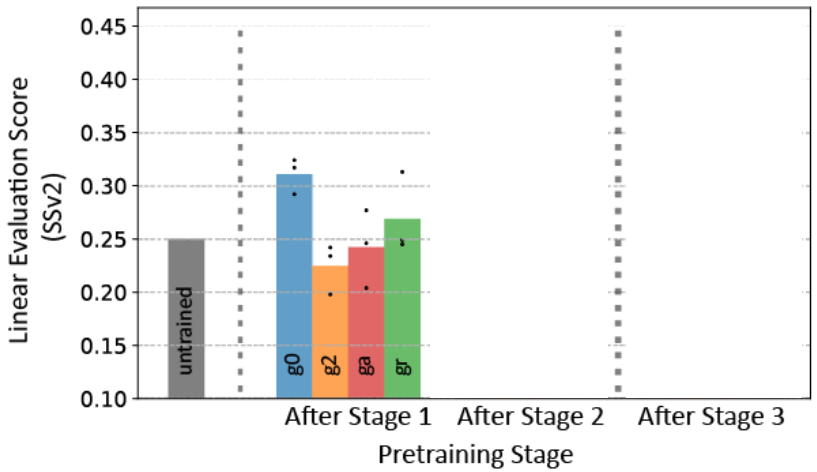
Curriculum learning with infant egocentric videos. NeurIPS (Spotlight paper), 2023.

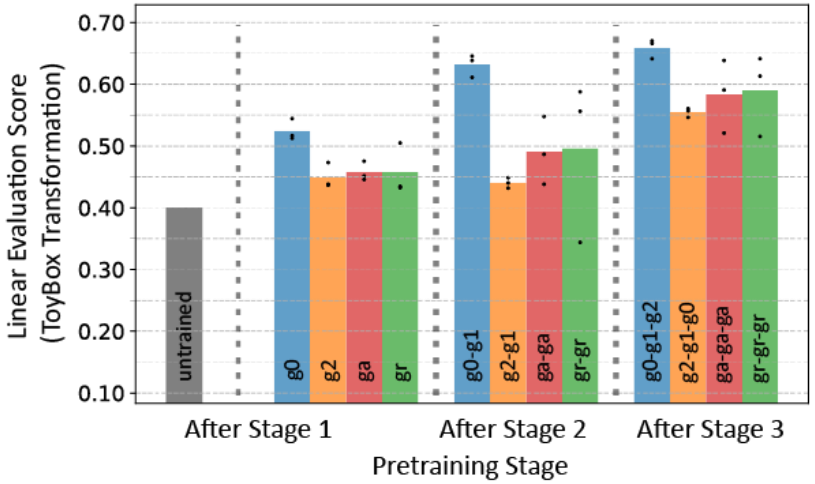
Curriculum learning with infant egocentric videos

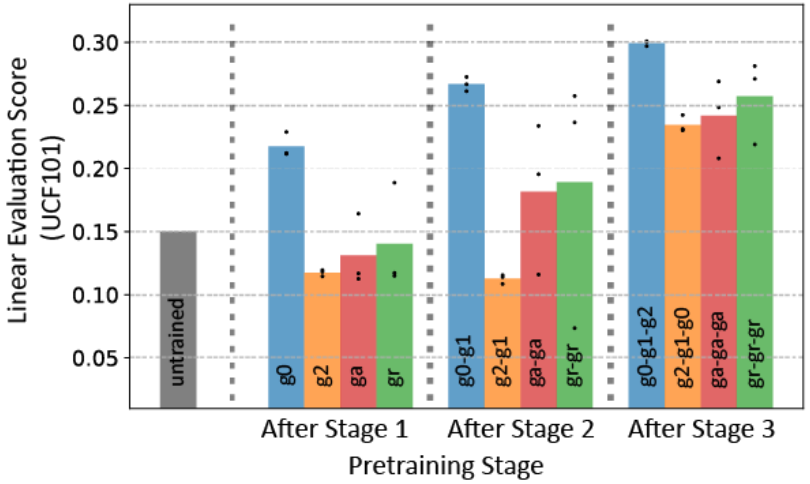
- The developmental curriculum outperforms all other curricula after any stage of training.











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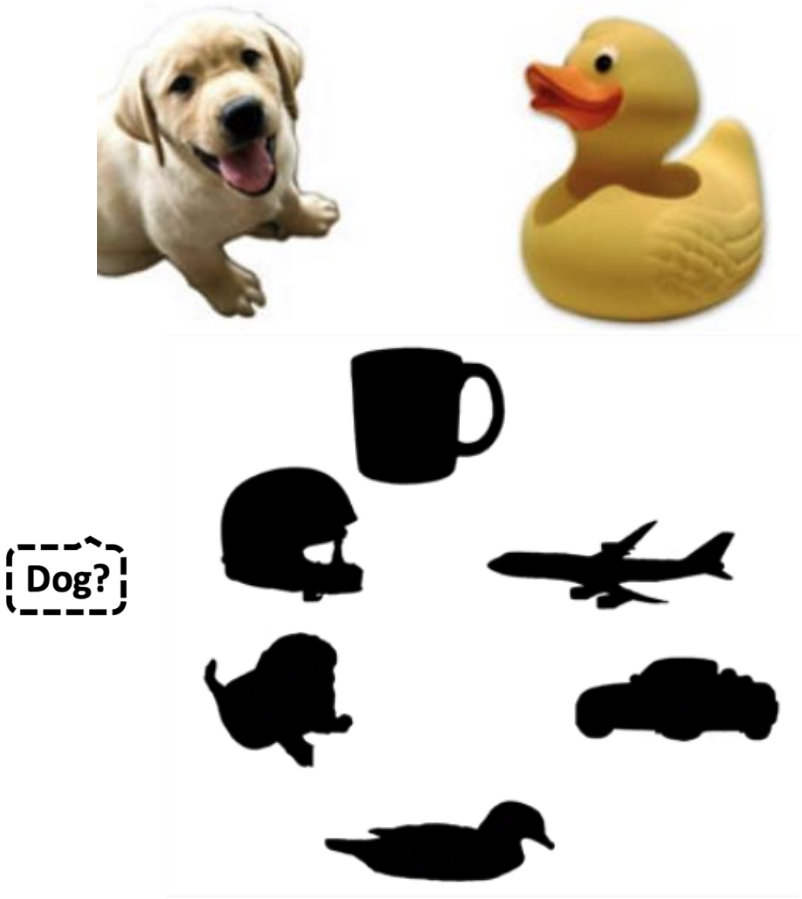
Curriculum learning with infant egocentric videos. NeurIPS (Spotlight paper), 2023.

ModelVsBaby: a Developmentally Motivated Benchmark for Out-of-Distribution Object Recognition

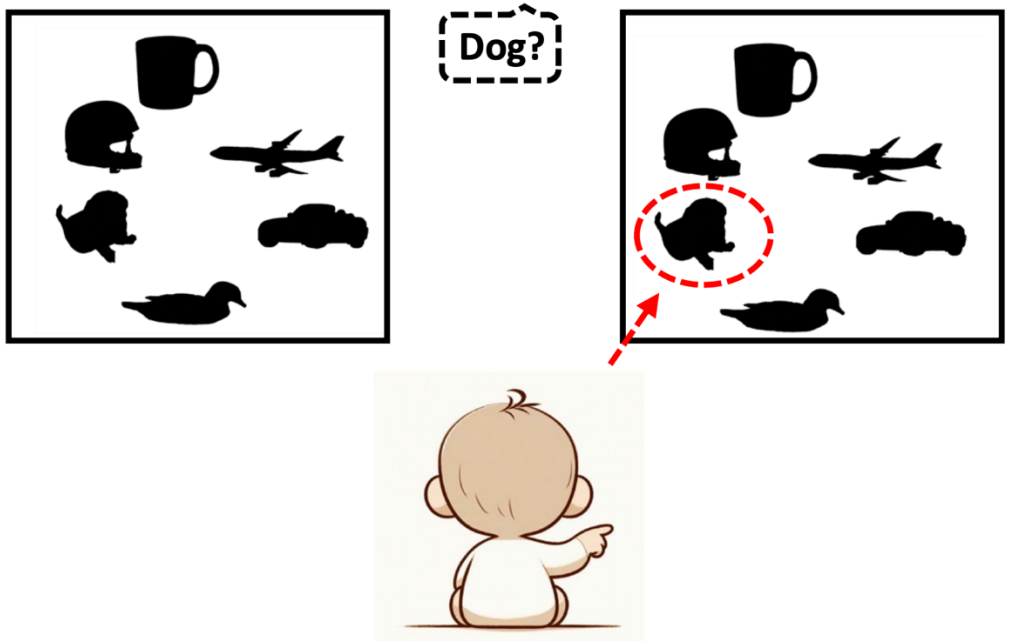
- How do we learn to categorize objects?
- Can a deep learning model do this?
  - Can a two-year old do this?
  - We created **ModelVsBaby**, the first comprehensive developmentally-motivated object recognition benchmark.



Saber Sheybani, Sahaj Maini, Aravind Dendukuri, Zoran Tiganj, Linda Smith. *ModelVsBaby: a Developmentally Motivated Benchmark of Out-of-Distribution Object Recognition,* OSF, 2024.

ModelVsBaby: a Developmentally Motivated Benchmark for Out-of-Distribution Object Recognition

Forced-choice image recognition task done by ~2-year-olds



8 common categories (5 instances from each)

5 conditions

Realistic **Silhouettes** **Geons** **Blurred** **Features**

Saber Sheybani, Sahaj Maini, Aravind Dendukuri, Zoran Tiganj, Linda Smith. *ModelVsBaby: a Developmentally Motivated Benchmark of Out-of-Distribution Object Recognition,* OSF, 2024.

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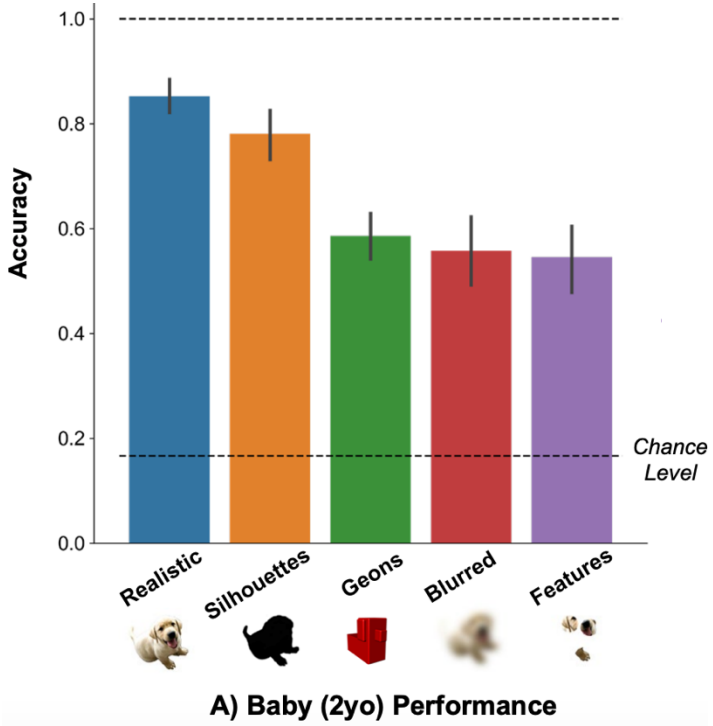

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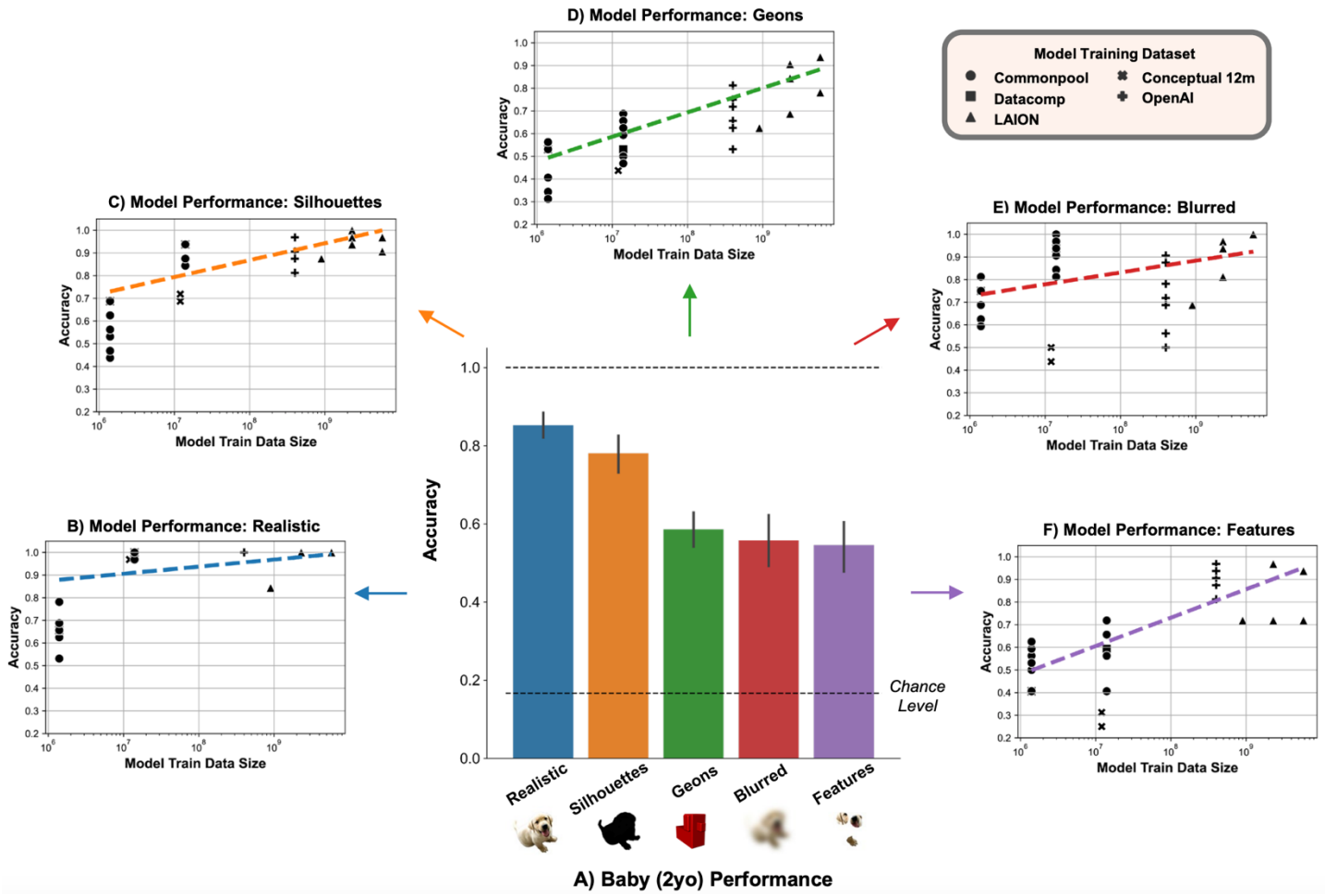

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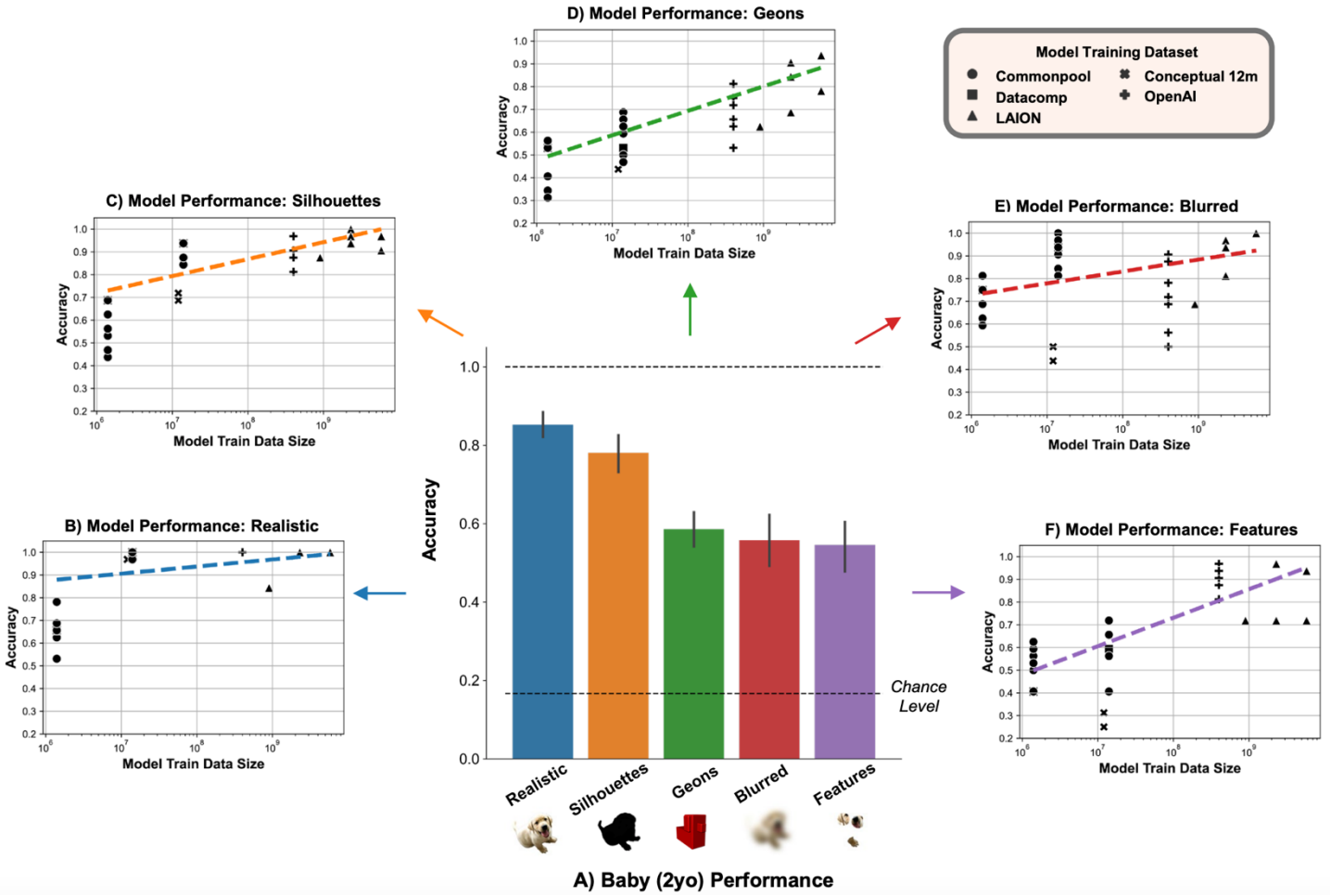

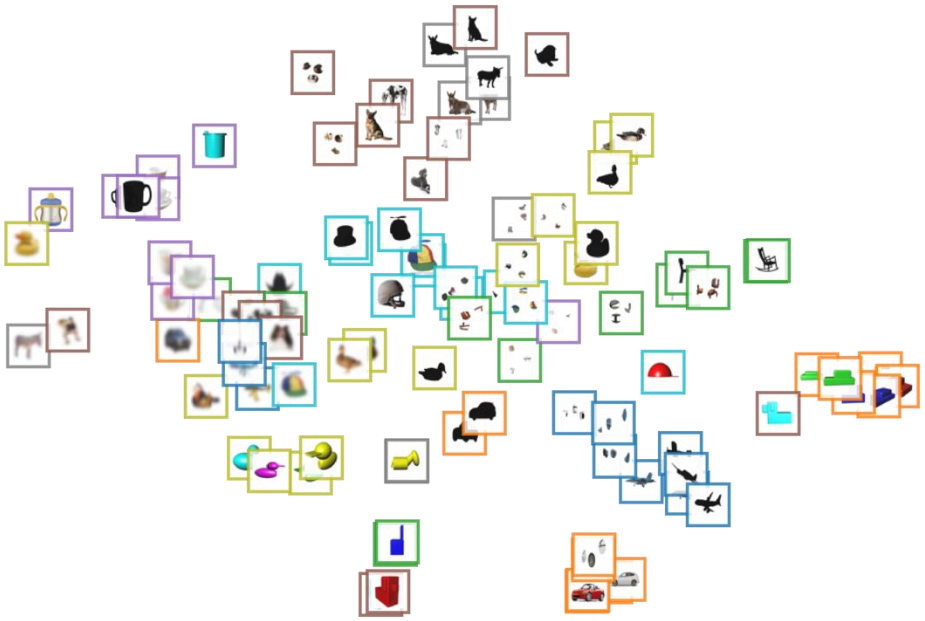

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Comparing Perceptual Judgments in Large Multimodal Models and Humans

Robert Nosofsky

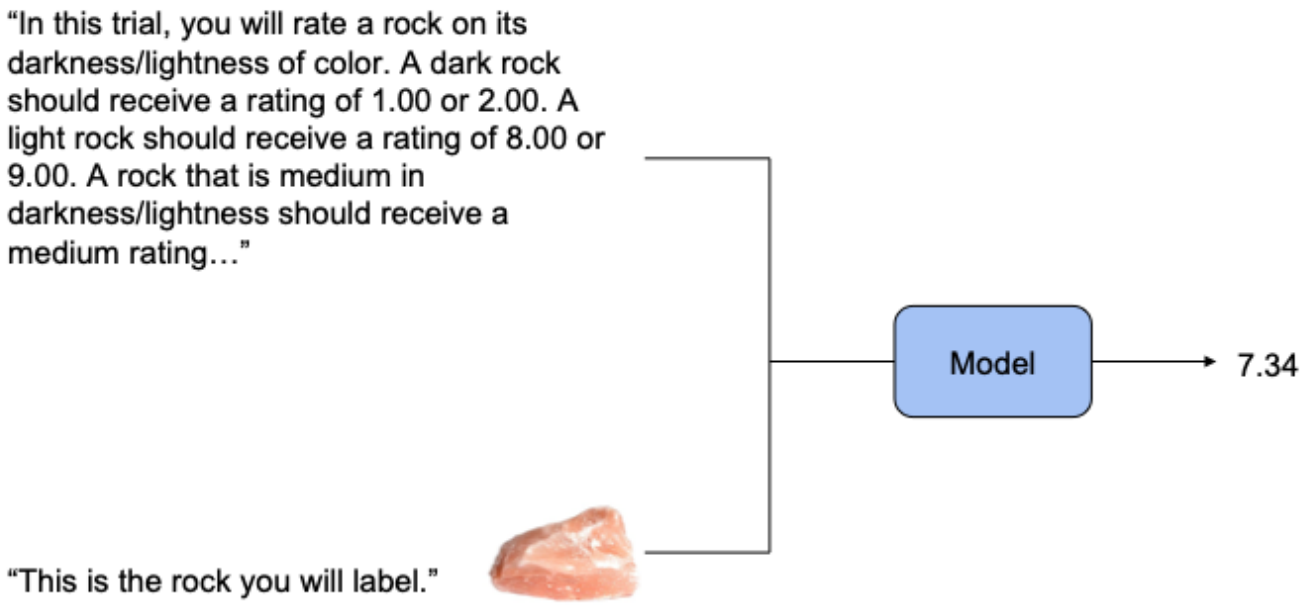
- How do we learn to categorize objects?
- How do people learn to differentiate these types of rocks?
  - chromaticity
  - darkness/lightness
  - disorganized/organized
  - dull/shiny
  - fine/coarse grain
  - red/green hue
  - smooth/rough
  - conchoidal fracture
  - pegmatitic structure
  - porphyritic texture
  - …

Granite

Diorite

Comparing Perceptual Judgments in Large Multimodal Models and Humans

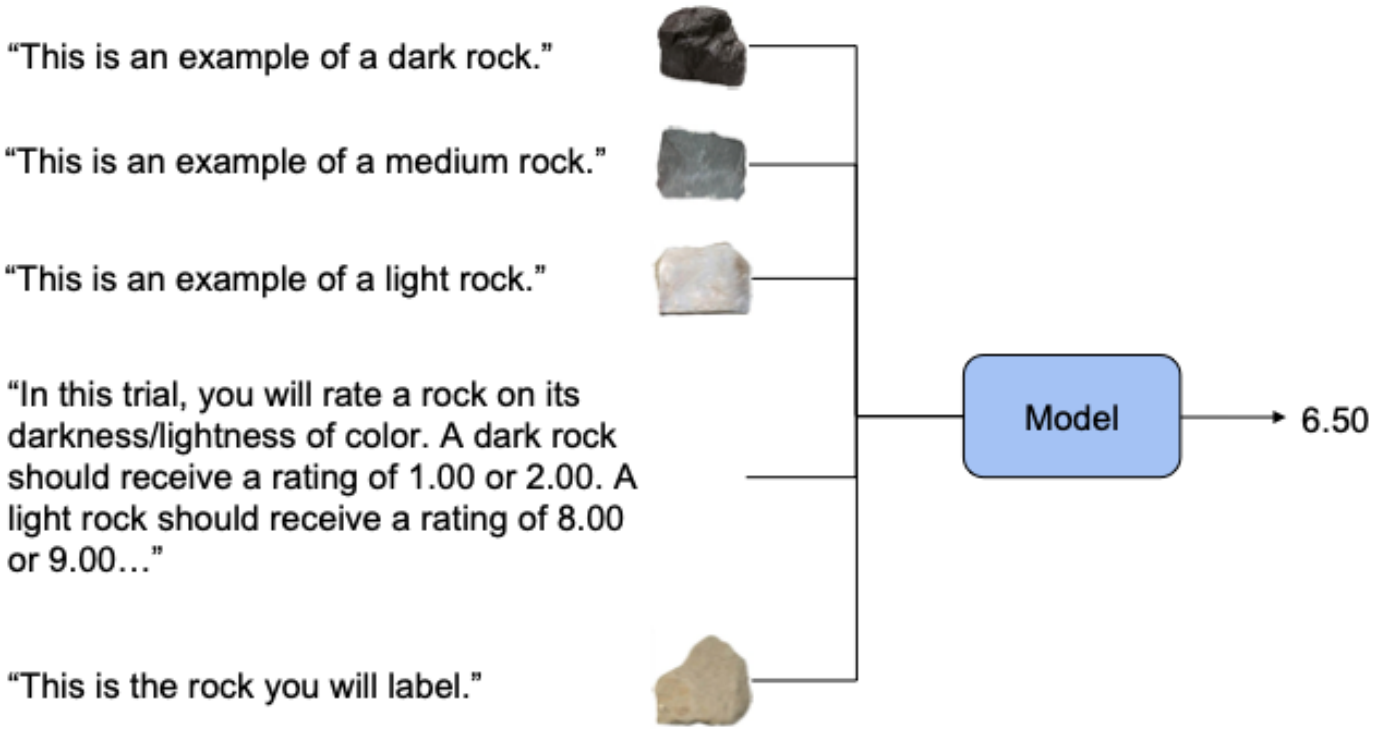
- We compared human perceptual judgments with perceptual judgments in large multimodal model.



Billy Dickson*, Sahaj Singh Maini*, Robert Nosofsky, Zoran Tiganj. *Comparing Perceptual Judgments in Large Multimodal Models and Humans,* PsychArXiv, 2024

Comparing Perceptual Judgments in Large Multimodal Models and Humans

- We compared human perce
  in large multimodal model.



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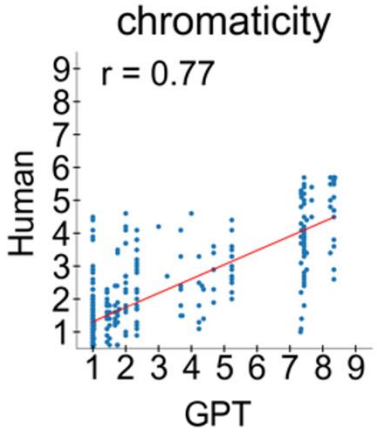
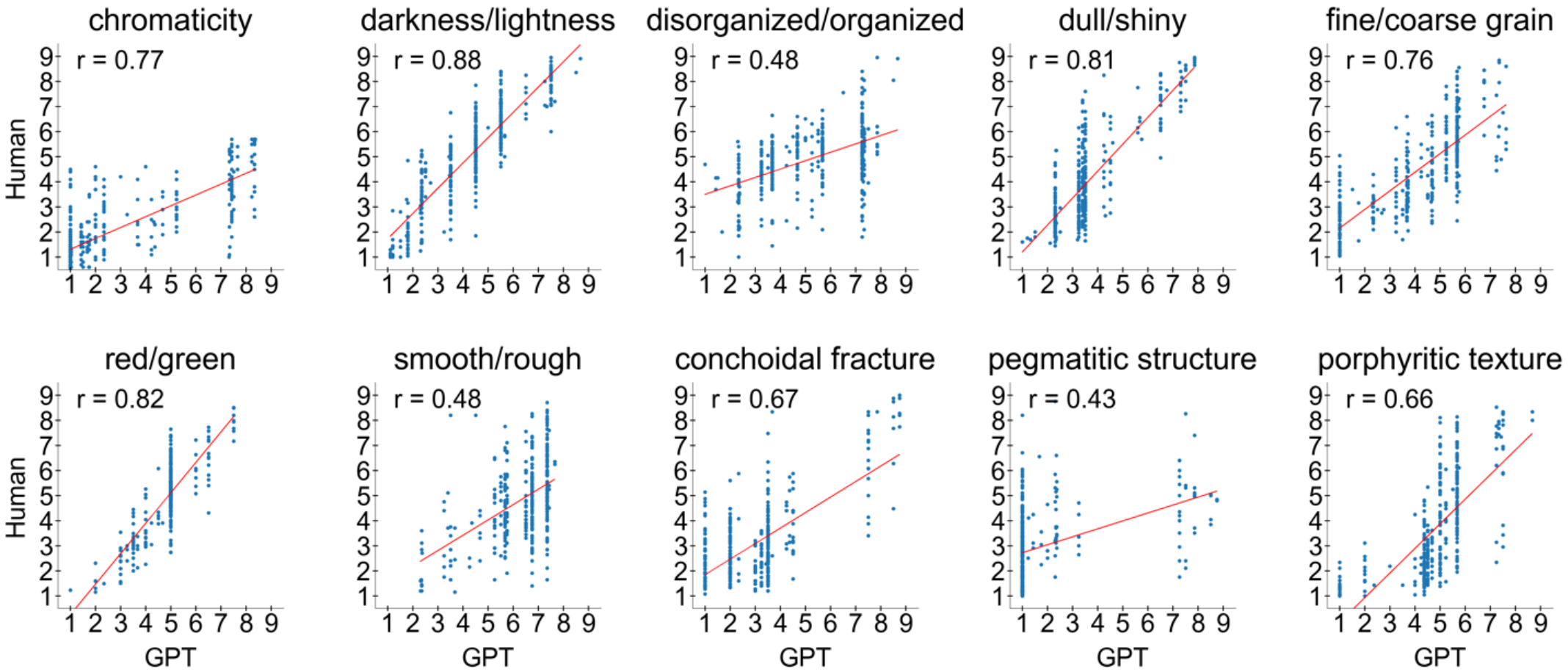


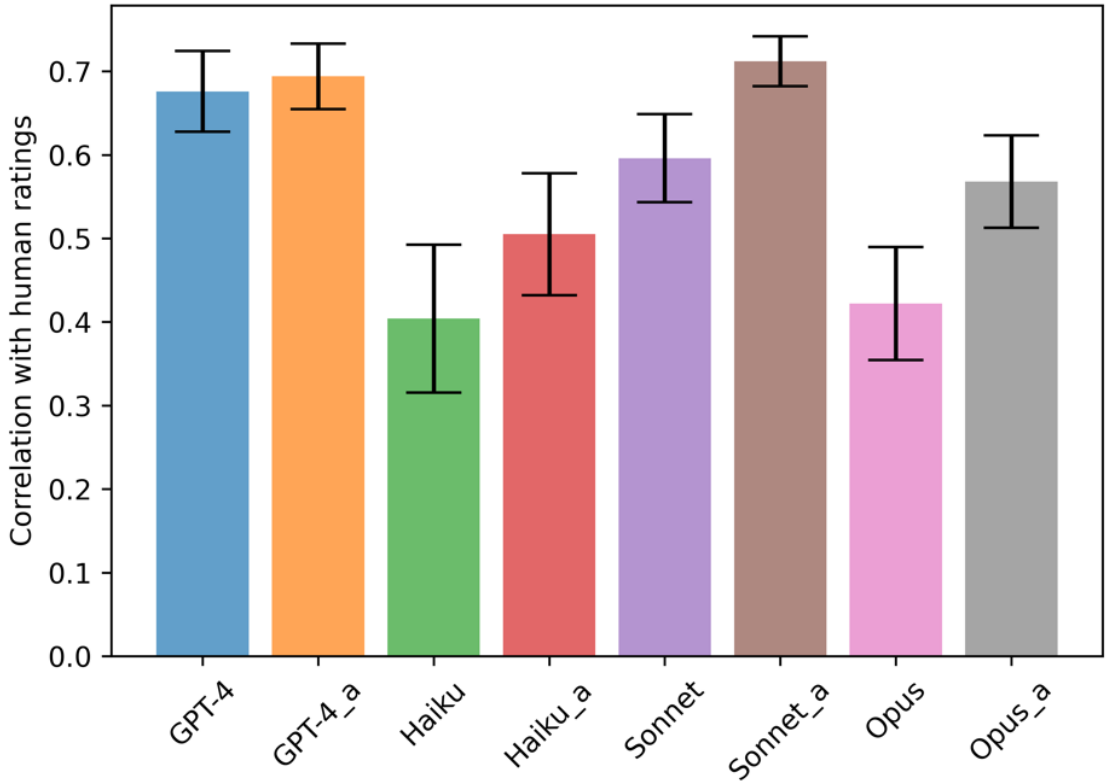
Billy Dickson*, Sahaj Singh Maini*, Robert Nosofsky, Zoran Tiganj. *Comparing Perceptual Judgments in Large Multimodal Models and Humans,* PsychArXiv, 2024

Comparing Perceptual Judgments in Large Multimodal Models and Humans



Billy Dickson*, Sahaj Singh Maini*, Robert Nosofsky, Zoran Tiganj. *Comparing Perceptual Judgments in Large Multimodal Models and Humans,* PsychArXiv, 2024

Comparing Perceptual Judgments in Large [illegible] nans



Billy Dickson*, Sahaj Singh Maini*, Robert Nosofsky, Zoran Tiganj. *Comparing Perceptual Judgments in Large Multimodal Models and Humans,* PsychArXiv, 2024

Incorporating a cognitive model for evidence accumulation into deep reinforcement learning agents

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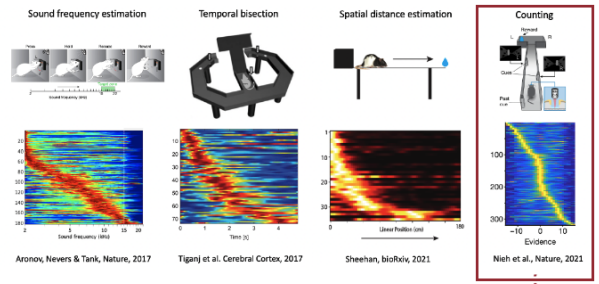
Motivation

The brain represents task-relevant physical and abstract variables as a cognitive map:



The **accumulating-towers task** has mice move down a virtual track, observing objects (towers) on the left and right, with the goal being to choose the side that had more towers.

Existing results indicate the existence of cells tuned to the **difference between number of towers**, which collectively tile the entire evidence axis [1].

Overview

Cognitive agents match performance of existing RNNs
Our cognitively-inspired agents are able to learn an evidence accumulation task as fast as LSTM and GRU-based agents, despite having much fewer trainable parameters and no trainable recurrent weights.

Our agents form similar neural activity to biological ones
Similar to hippocampal recordings in mice, the neural activity generated by our agents reflects a low-dimensional ordered representation of evidence.

Our agents can be trained end-to-end using RL
Error signals can be backpropagated through our evidence accumulation module, allowing a convolutional network to be trained to encode pixel inputs into an appropriate representation.

Our approach can generalize to other task-relevant variables
The deep network learns to modulate the speed of sequential activation with the amount of change in a task-relevant variable, turning the temporal map into a map representing that variable.

Evidence Accumulation

The Laplace transform is implemented via an RNN with analytically-derived weights [2]:

$$\mathbf{F}_{s;t} = \mathbf{W}\mathbf{F}_{s;t-1} + f_t$$

(recurrent weight matrix) $\mathrm{diag}(e^{-\alpha(t)s\Delta t})$

$s = [1.6 \quad 1.366 \quad 1.167 \quad \cdots \quad 0.08]$ (intrinsic decay rate – log-spaced)

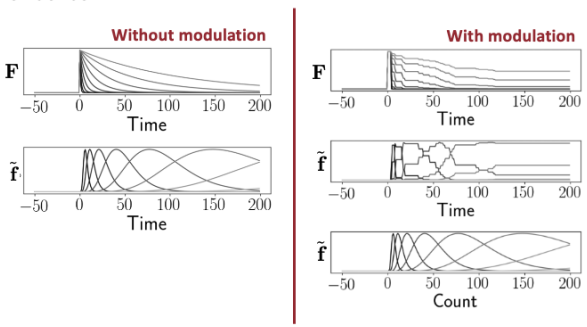
$\alpha = \frac{\Delta n}{\Delta t}$ (change in numerosity)

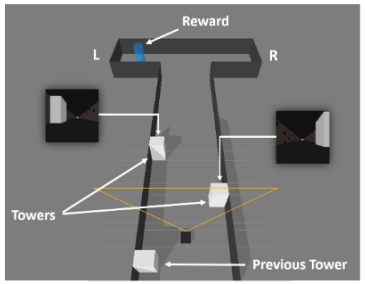
$\delta(0)$ [1 0 0 0 0 ... 0]

$$\tilde{\mathbf{f}}_{n^*;n} = \mathbf{L}_k^{-1}\mathbf{F}_{s;t}$$

$\frac{(-1)^k}{k!} s^{k+1} \frac{d^k}{ds^k}$ (Post's inversion formula)

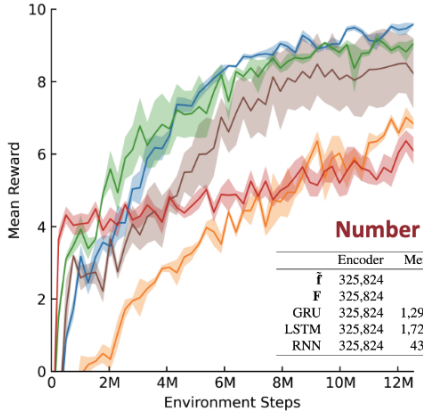
Modulation of α is reflected in similar modulation of the decay rate of $\mathbf{F}$ neurons. This leads to $\tilde{\mathbf{f}}$ neurons tuned to amounts of evidence:





Results

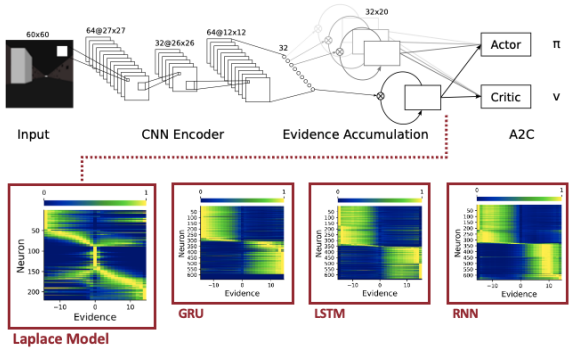
Loss Curve



Number of Parameters

	Encoder	Memory	RL	Total
$\tilde{\mathbf{f}}$	325,824	0	823,044	1,148,868
$\mathbf{F}$	325,824	0	823,044	1,148,868
GRU	325,824	1,294,080	823,044	2,442,948
LSTM	325,824	1,725,440	823,044	2,874,308
RNN	325,824	431,360	823,044	1,580,228

Neural Activity



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1. R. J. Low, S. Lewallen, D. Aronov, R. Nevers, and D. W. Tank. Probing variability in a cognitive map using manifold inference from neural dynamics. BioRxiv, page 418939, 2018.

2. M. W. Howard, C. J. MacDonald, Z. Tiganj, K. H. Shankar, Q. Du, M. E. Hasselmo, and H. Eichenbaum. A unified mathematical framework for coding time, space, and sequences in the hippocampal region. *Journal of Neuroscience*, 34(13):4692–707, 2014.

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Lastly

- For any questions, comments and concerns, please do not hesitate to contact me or our AIs.
- Thank you for taking the course, I hope you will learn a lot and have fun doing so!